



T30 Module Introduction

A DJI T30 agricultural drone is shown in flight, spraying a green field. The drone is white and black, with orange propellers. It is carrying a large white tank and is spraying a fine mist of liquid onto the field below. The background is a lush green field with some trees in the distance.

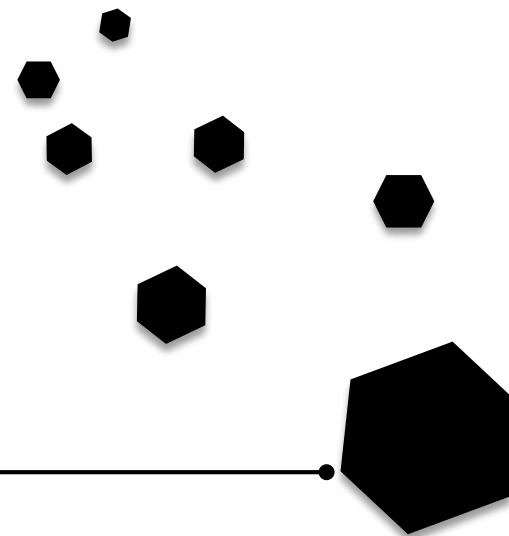
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01

Part One

T30 Aircraft Modules



Cable Distribution Module

The **cable distribution module** uses a fully enclosed structure for added durability. Three layers of protection over key components offer an overall rated water resistance of IP67 against pesticides, dust, fertilizers, and corrosion.



Aerial-Electronics
System Module



Upward Radar



Spraying Module

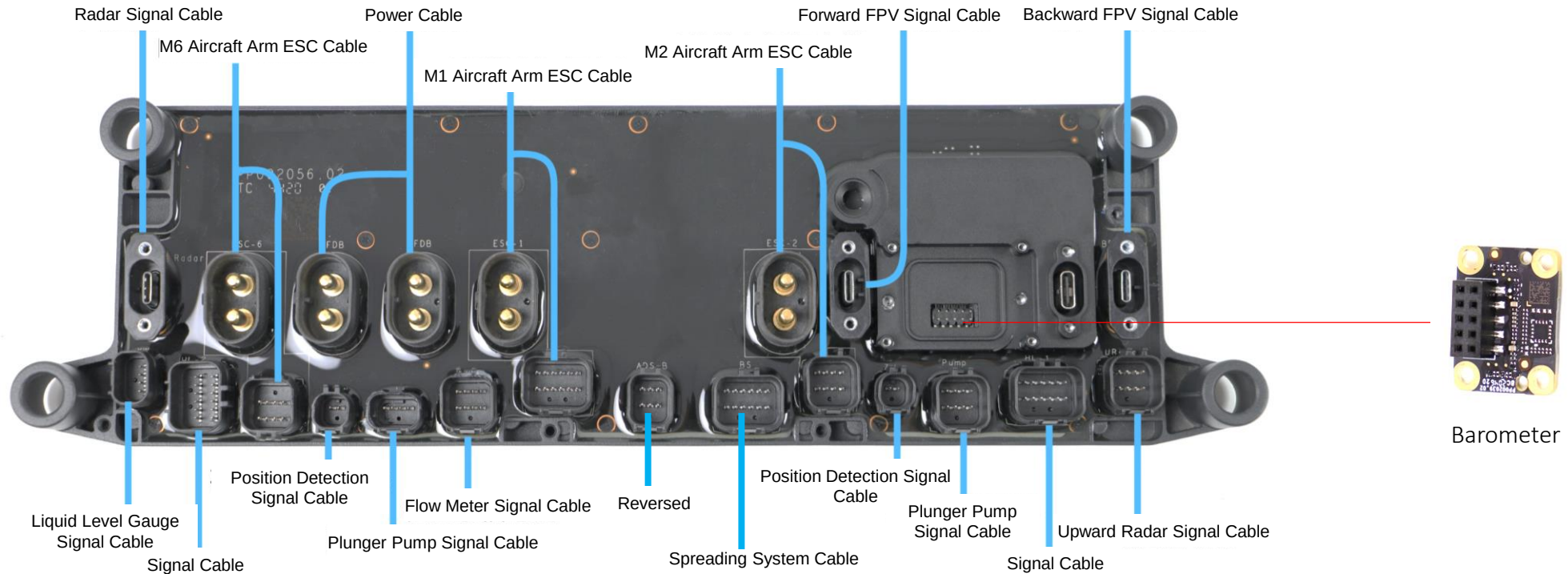


RF Module



Cable Distribution board

Cable Distribution Board



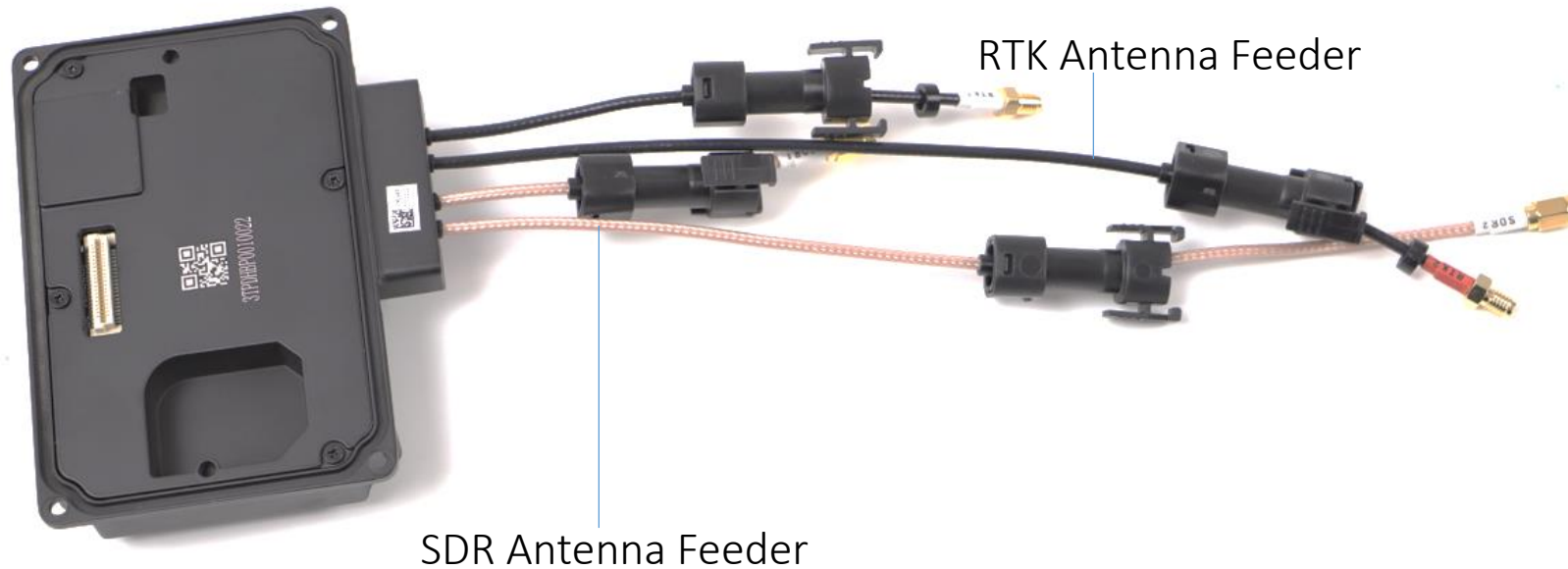
Cable Distribution Board: Summarizes the information of each module and contains a barometer. The barometer is mainly used to detect the altitude of the aircraft.

Aerial-Electronics System Module



Aerial-Electronics System Module: Flight control system of the aircraft, which processes the information of each sensor and feeds it back to the propulsion system for adjustment, and includes the compass.

RF Module



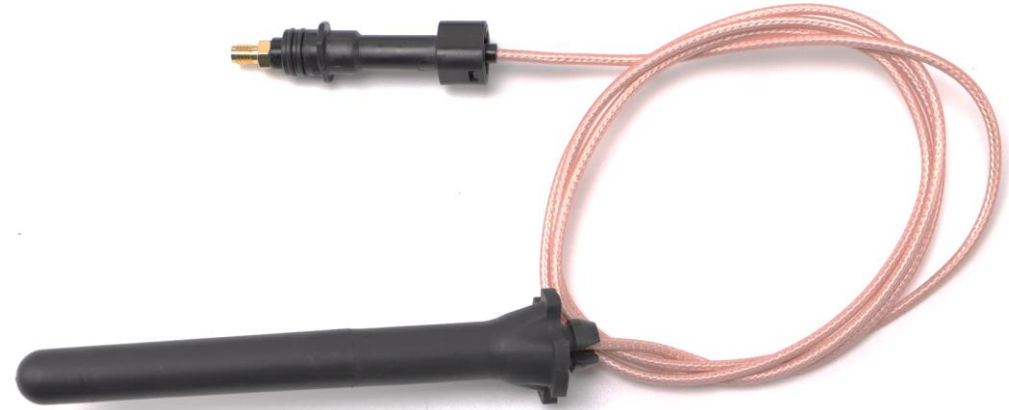
RF Module: Processes the image transmission of the aircraft, communicate with the remote controller and transmit related information including image transmission. At the same time, it also processes GPS/RTK information for positioning;

Antennas



RTK Antenna

The **RTK Antenna module** provides information such as position, direction, and speed of the aircraft for the flight controller system to assist with the stability of aircraft navigation and the accuracy of operation



SDR Antenna

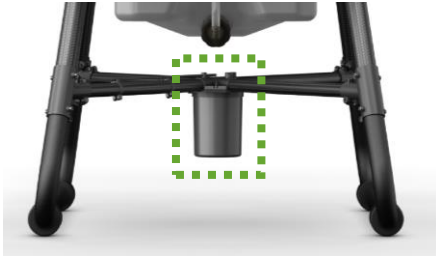
The **SDR Antenna module** for image signal and control signal transmitting and receiving between aircraft and remote controller.

Spraying Module

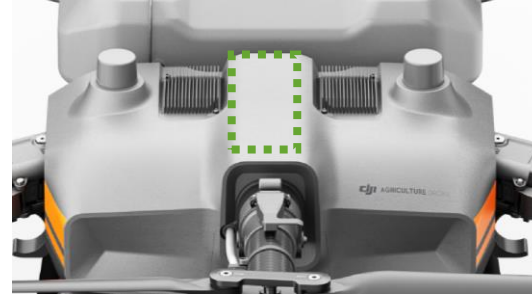


Spraying Module: Control system of the aircraft's core spray function, responsible for controlling the delivery pump and the amount of pesticide sprayed.

Radar



Omnidirectional Digital Radar

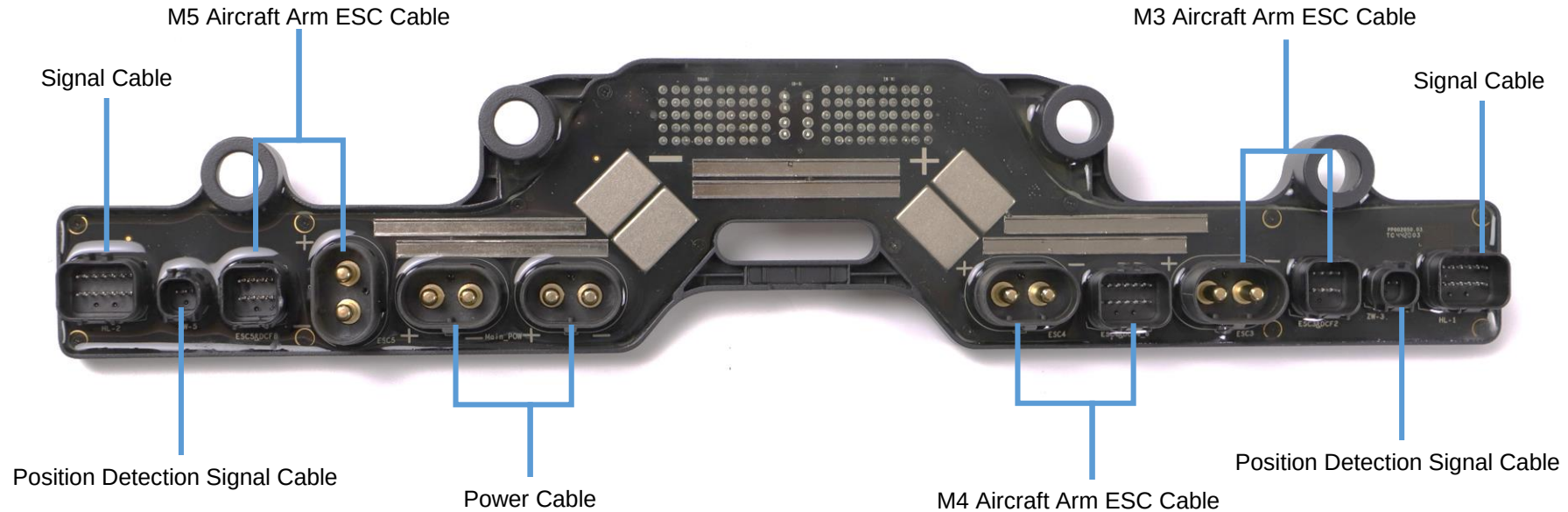


Upward Radar

Spherical Perception Radar System includes: **Omnidirectional Digital Radar + Upward Radar**

Supports functions such as terrain following, obstacle sensing, and obstacle circumvention, comprehensively ensuring operational safety day and night in different weather; additionally, it is dustproof and free of lighting interference.

Power Distribution Module



The **Power Distribution Module** and the cable distribution module have fully enclosed potting protection.

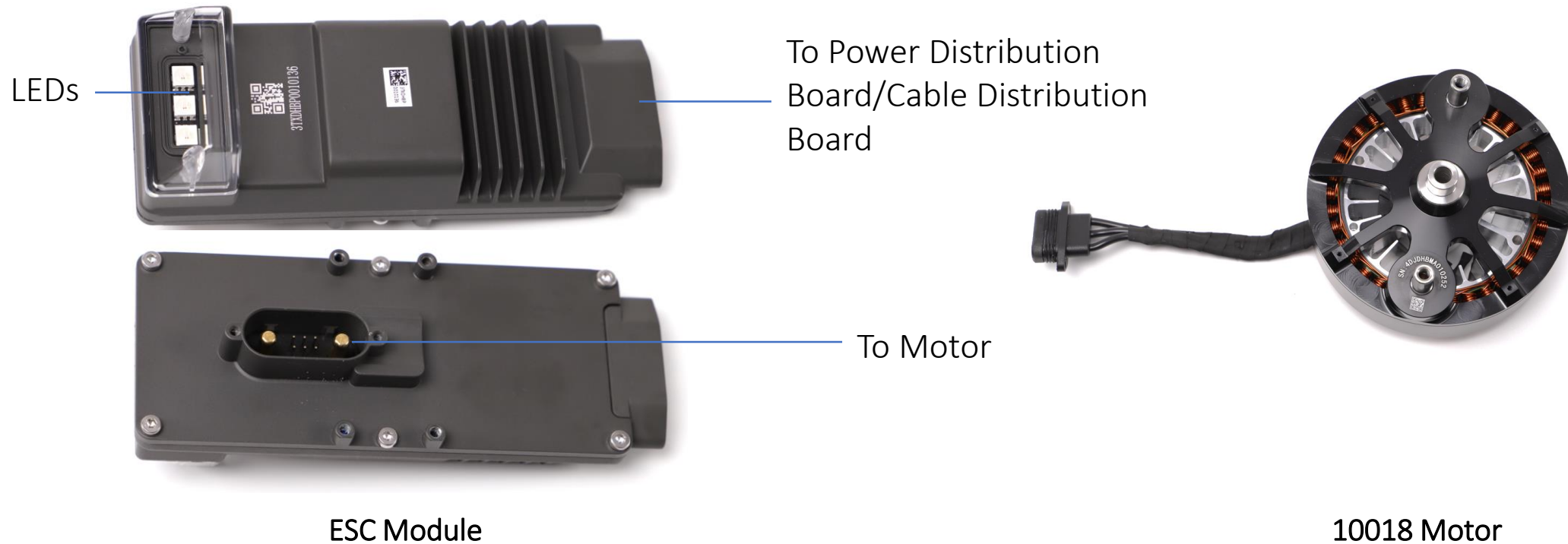


FPV Camera Module



T30 Equipped with dual **FPV cameras**, they provide clear front and rear views and enables the user to check the flight status without turning the aircraft around. Furthermore, its bright spotlight doubles the drone's night vision capabilities, making it easier to operate at night.

ESC/Motor



The **ESC module** is responsible for converting the throttle information into the motor speed information, controlling the motor rotation, reporting the motor speed to the flight controller, and providing functions such as overcurrent protection.

The **Motor** is the executive part of the aircraft, which drives the propellers to provide the upward propulsion.

Plunger Pump/Electromagnetic Valves



Plunger Pump



Electromagnetic Exhaust Valves

Spraying System: Equipped with new plunger pumps, 16 sprinklers, and 8 electromagnetic exhaust valves (each of which can independently control two nozzles), the spraying system provides even and accurate spraying so that users can save liquid and reduce operating costs.

Delivery Pump: Actuator of the aircraft spraying system to spray pesticides through rotation. The T30 delivery pump uses a plunger pump design, and the T10 uses a diaphragm pump design.

Electromagnetic Exhaust Valves: Adjust the air in the water pipes, and they can independently switch any of the nozzles for spraying.

Flow Meter/Liquid Level Gauge



Flow Meter Module



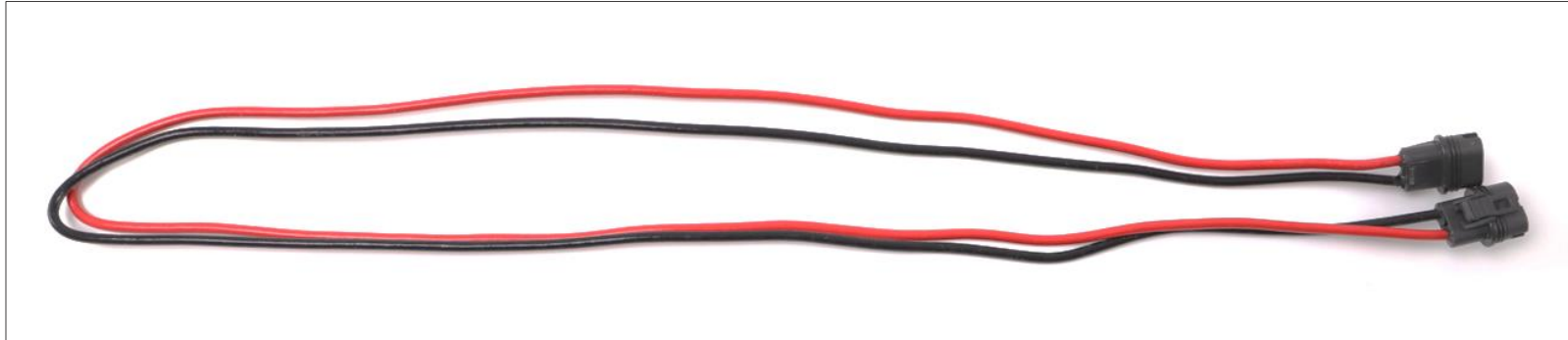
Liquid Level Gauge

2-Channel Flow Meter: Core component of high-precision spraying, responsible for detecting the liquid fertilizer sprayed by the delivery pump and for accurately detecting the flow rate.

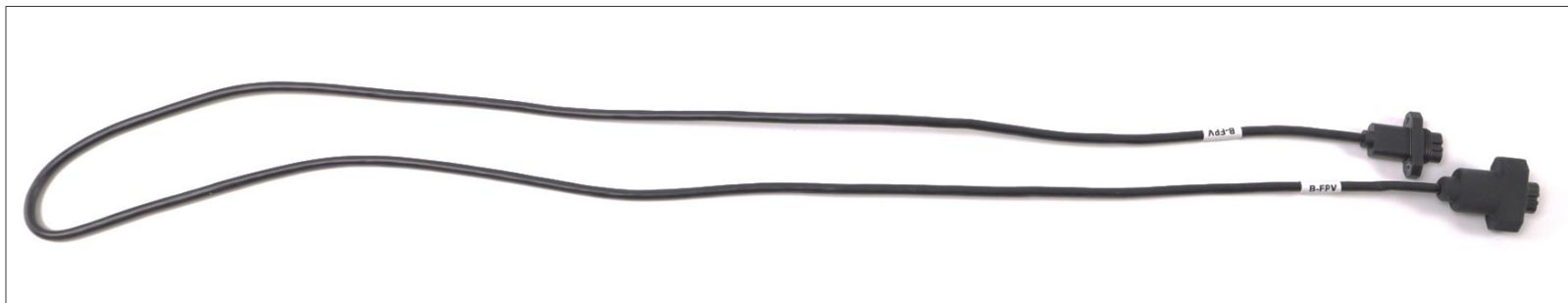
Continuous Liquid Level Gauge: Obtain the amount of the liquid and make alerts when the tank is empty. The 2-channel electromagnetic flow meter and continuous liquid level gauge make measurements more accurate than before. The new Smart Supply Reminder calculates the remaining liquid amount to help users manage spraying operations.

Cables

Power Cable

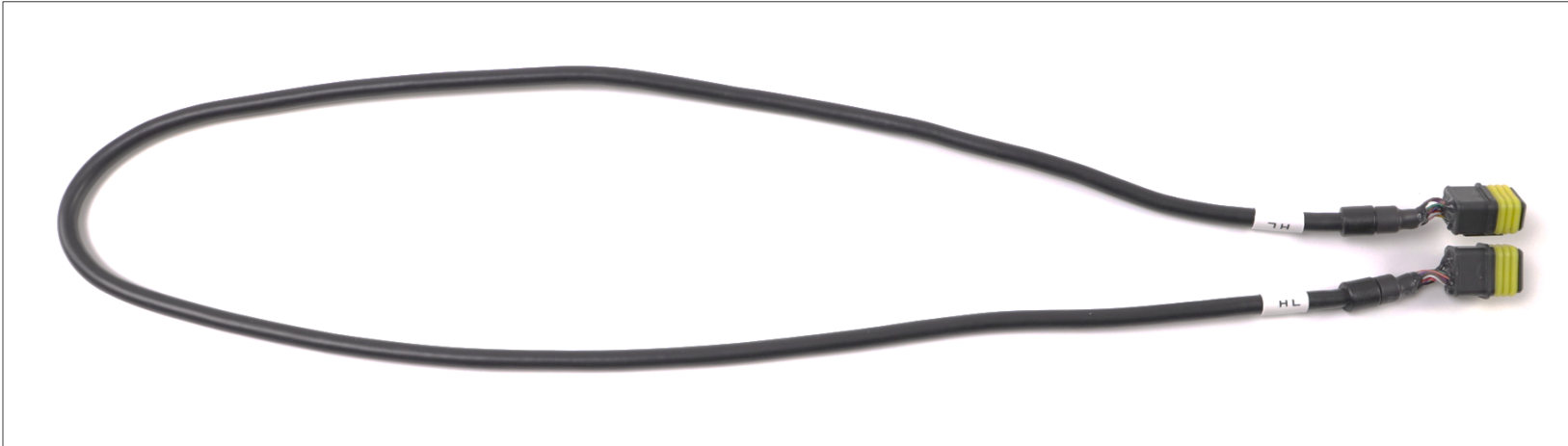


Rear FPV Signal Cable

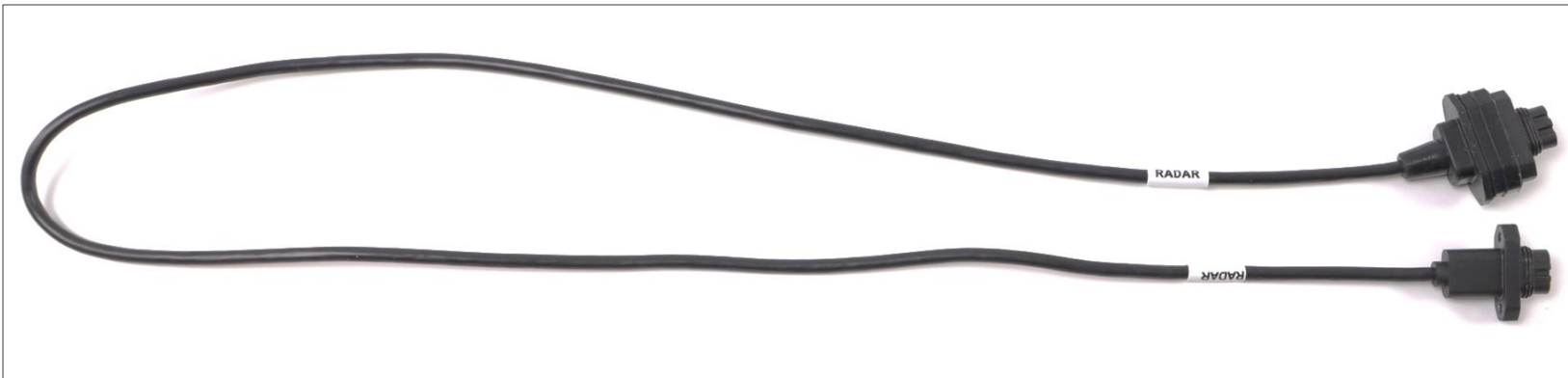


Cables

Signal Cable



Radar Signal Cable



Cables

Liquid Level Gauge Signal Cable



Flow Meter Signal Cable



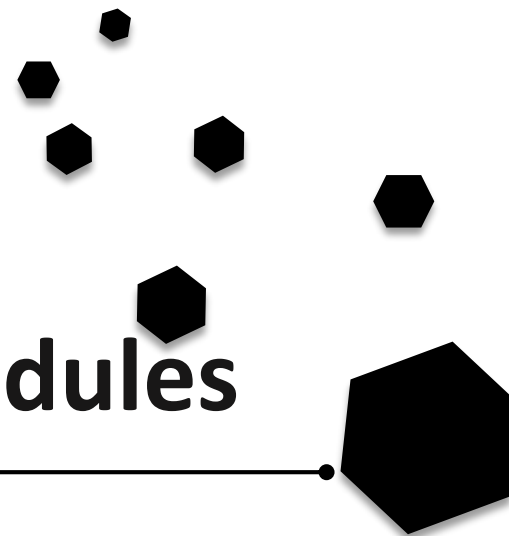
Front FPV Signal Cable



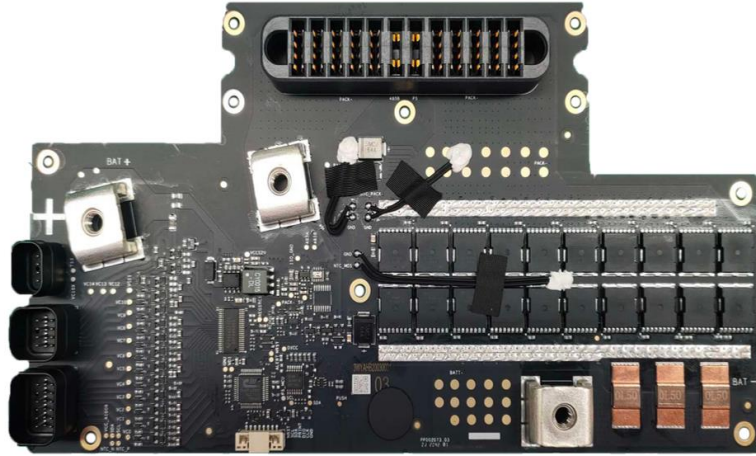
02

Part Two

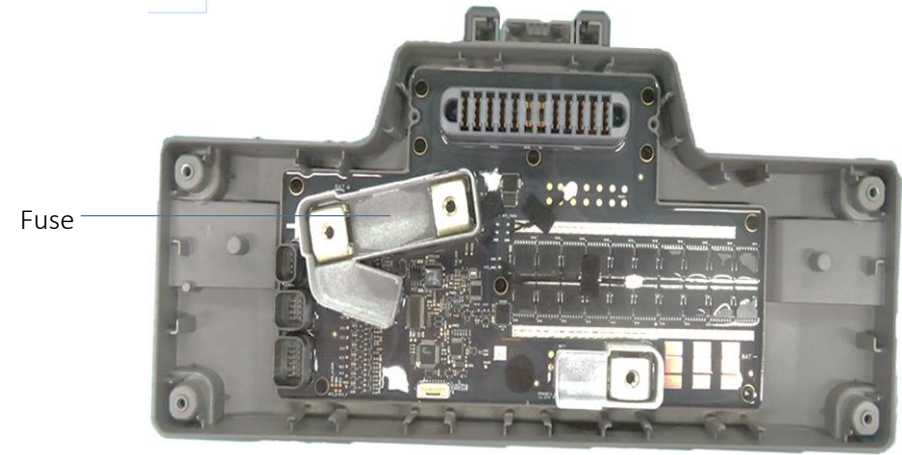
Battery and Charger Modules



Battery Main Board/Battery Upper Shell



Battery Main Board



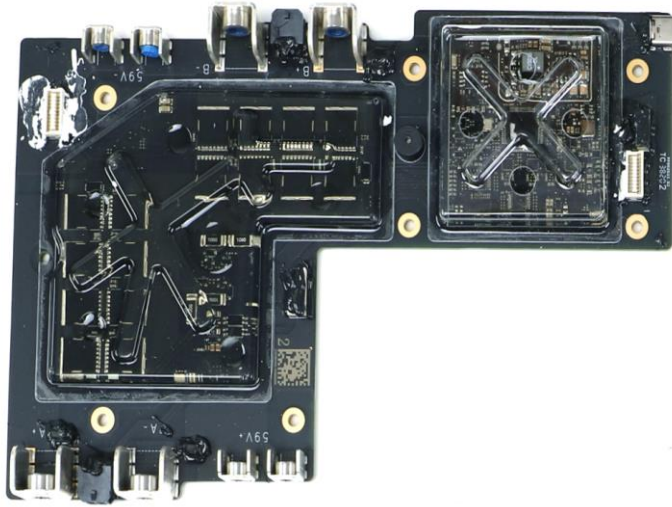
Battery Upper Shell Module

(Glue filling process to completely fix the battery main board in the upper shell)

Battery Main Board: Communicates with the battery station or aircraft, and manages and calculates the capacity of the battery cell module. The LED can display the battery status.

Upper Shell Module (Including battery adaptive board): Connects the battery cell, and connects the battery main board to the external charger and aircraft.

Battery Station main Board/LED Board/Power Button Module



Battery Station Main Board: Communicates with the power module and battery, manages the battery charging current, and outputs the LED signal to control the LED display. The management board firmware can be updated by connecting to the computer via Type-C port or on the remote controller. The log can be exported through the computer and imported via the remote controller.



LED board: Connects the management board and the system switch respectively, and transmits the switch signal to the management board through the LED board. The LED on the LED board is used to display the battery charging status and the status of the battery station.



Power Button Module: Controls powering on or off of the battery station.

Battery Charging Cable/Power Module/Adaptive Board



T30 Battery Charging Cable: An adaptive cable between the main board and battery, providing charging voltage and communication



Power Module Adaptive Board: Transfers power between the 4850 module and the main board. The power output of the power module is transferred to the main board through the 4850 module.

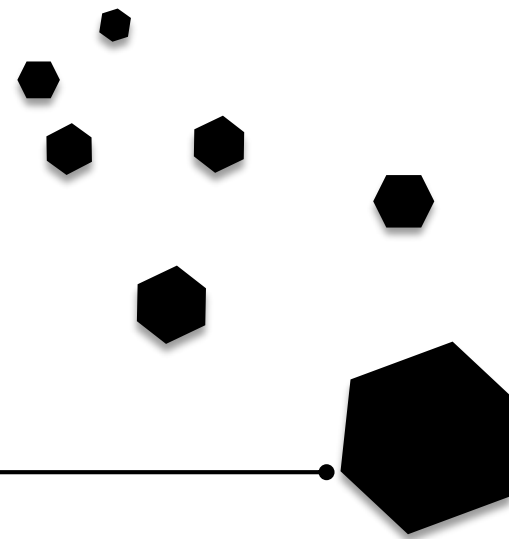


Power Module: The 220V electric power is converted into 59V direct current to charge the battery; also communicates with the main board. The maximum output current of a single power module can reach 65A.

03

Part Three

Spreading System



Control Module/Servo Module/Reduction Gearbox



Control Module

Core module of the spreading system.



Servo Module

Controls the change of the hopper gate angle and the opening/closing of the hopper gate.



Reduction Gearbox

Drives the stirring device to spin, brings the materials to the hopper gate, and spreads them out of the hopper gate.

Weight Sensor/Weight Collection Board



Weight Sensor

Responsible for collecting the weight of the tank.



Weight Collection Board

Receives the data from the weight sensor and calculates the weight.

THANK YOU

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